

20 Magnetic fields

A · 2026 examination · Concept guide and original exam-style practice

Understand the concept

A current-carrying conductor in a magnetic field experiences $F = BIL \sin \theta$. A moving charge experiences $F = Bqv \sin \theta$; the force is perpendicular to velocity and does no work. Faraday's law relates induced e.m.f. to the rate of change of magnetic flux linkage, with Lenz's law setting the opposing direction.

Worked example: follow the reasoning

A 0.20 m wire carries 3 A at right angles to $B = 0.50$ T: $F = 0.50 \times 3 \times 0.20 = 0.30$ N.

Quick check · multiple choice

A Magnetic force on a moving charge at right angles to B is what?

[1]

- A Bq/v
- B Bv/q
- C qv/B
- D Bqv

Write your choice: _____

Structured practice

Answer in the space provided. Show working and units where relevant.

1 A 50-turn coil's flux per turn changes from 0.020 to 0.0050 Wb in 0.10 s. Find mean induced e.m.f. magnitude.

[3]

Final answer: _____

Answers and explanation

Quick check: D. Bqv

Question 1 [3 marks]

$\Delta(N\Phi) = 50(0.020 - 0.0050) = 0.75 \text{ Wb-turn [1]}$; $\text{abs}(\epsilon) = \Delta(N\Phi)/\Delta t \text{ [1]} = 7.5 \text{ V [1]}$.

Self-assessment

Compare each credited step with your own answer. Fix the first reasoning or calculation error; then retry without looking at the solution.

Syllabus: Physics 9702, 2025–2027. <https://www.cambridgeinternational.org/Images/664565-2025-2027-syllabus.pdf>

Original FagusED teaching material. Question layout and mark presentation informed by the May/June 2026 papers supplied by the user. The practice question is newly written and is not a Cambridge past-paper question.